** OFFICIAL UNIVERSITY PHYSICS MASTER FORMULA & PHYSICAL CONSTANTS SHEET **

PHYSICS I: MECHANICS, OSCILLATIONS, WAVES & THERMODYNAMICS

Comprehensive Equation Compendium • Standard SI Units & Exact Constants • Learning Hub Official Master Repository

SECTION 0: FUNDAMENTAL PHYSICAL CONSTANTS & STANDARD MATERIAL VALUES

Quantity	Sym	Value	SI Unit
Standard Gravity	g	**9.80665 ≈ 9.80**	m/s² (N/kg)
Universal Gravitation	G	**6.6743×10^{-11}**	N·m²/kg²
Universal Gas Constant	R	**8.31446** (0.08206)	J/(mol·K)
Boltzmann Constant	k_B	**1.38065×10^{-23}**	J/K
Avogadro's Number	N_A	**6.02214×10^{23}**	mol^{-1}
Stefan-Boltzmann	σ	**5.67037×10^{-8}**	W/(m²·K⁴)
Standard Atm. Pressure	P_{atm}	**1.01325×10^{5}**	Pa (760 Torr)
Hearing Threshold (0 dB)	I_0	**1.00×10^{-12}**	W/m²
Speed of Sound (Air, 20°C)	v_{sound}	**343** (331.4 + 0.6T_C)	m/s

Quantity	Sym	Value	SI Unit
Density of Pure Water (4°C)	ρ_{water}	**$1.000 \times 10^3 = 1000$**	kg/m³ (1 g/cm³)
Density of Mercury (0°C)	ρ_{Hg}	**$13.595 \times 10^3 \approx 13.6 \times 10^3$**	kg/m³
Density of Air (20°C, 1 atm)	ρ_{air}	**1.204**	kg/m³
Specific Heat of Water	c_{water}	**4186** (1.000 cal/g·°C)	J/(kg·K)
Specific Heat: Ice / Steam	c_{ice}/c_{st}	**2090 / 2010**	J/(kg·K)
Latent Heat of Fusion (Ice)	L_f	**3.33×10^5** (79.7 cal/g)	J/kg
Latent Heat of Vap. (Water)	L_v	**$2.256 \times 10^6 \approx 2.26 \times 10^6$**	J/kg
Molar Heat Capacity (Mono)	C_v, C_p, γ	$\frac{3}{2}R$, $\frac{5}{2}R$, **1.67**	J/(mol·K), γ=1.67
Molar Heat Capacity (Dia)	C_v, C_p, γ	$\frac{5}{2}R$, $\frac{7}{2}R$, **1.40**	J/(mol·K), γ=1.40

SECTION 1: KINEMATICS, PROJECTILE & RELATIVE MOTION

1.1 One-Dimensional Constant Acceleration Equations

$v = v_0 + at$

$x = x_0 + v_0 t + \frac{1}{2}at^2$

$x = x_0 + vt - \frac{1}{2}at^2$

$v^2 = v_0^2 + 2a(x - x_0)$

$x - x_0 = \frac{1}{2}(v_0 + v)t$

$v(t) = \frac{dx}{dt}$

$a(t) = \frac{dv}{dt} = \frac{d^2x}{dt^2} = v\frac{dv}{dx}$

1.2 Two-Dimensional Projectile Motion (Flat Ground, Neglecting Air Resistance)

$v_{0x} = v_0 \cos\theta$

$v_{0y} = v_0 \sin\theta$

$x(t) = (v_0 \cos\theta)t$

$y(t) = (v_0 \sin\theta)t - \frac{1}{2}gt^2$

$v_x = v_{0x} = \text{const}$

$v_y(t) = v_{0y} - gt$

$v = \sqrt{v_x^2 + v_y^2}$

$\tan\alpha = \frac{v_y}{v_x}$

Time of Flight: $T = \frac{2v_0 \sin\theta}{g}$

Peak Height: $H = \frac{v_0^2 \sin^2\theta}{2g}$

Horizontal Range: $R = \frac{v_0^2 \sin(2\theta)}{g}$

Trajectory: $y(x) = (\tan\theta)x - \frac{g}{2v_0^2 \cos^2\theta}x^2$

1.3 Circular Kinematics & Relative Motion

$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega}$

$f = \frac{1}{T} = \frac{\omega}{2\pi}$

$s = r\theta$

$v = r\omega$

$a_t = r\alpha$

$a_c = \frac{v^2}{r} = \omega^2 r = \frac{4\pi^2 r}{T^2}$

$a_{total} = \sqrt{a_c^2 + a_t^2}$

$\mathbf{r}_{P/A} = \mathbf{r}_{P/B} + \mathbf{r}_{B/A}$

$\mathbf{v}_{P/A} = \mathbf{v}_{P/B} + \mathbf{v}_{B/A}$

SECTION 2: NEWTON'S LAWS, FRICTION & CIRCULAR DYNAMICS

2.1 Laws of Dynamics & Gravity

$\sum \mathbf{F} = m\mathbf{a} = \frac{d\mathbf{p}}{dt}$

$\mathbf{F}_{12} = -\mathbf{F}_{21}$

$F_g = mg$

Universal Gravitation: $F_G = \frac{G m_1 m_2}{r^2}$

Incline: $F_{\parallel} = mg \sin\theta$

Incline: $F_{\perp} = N = mg \cos\theta$

Apparent Weight (Elevator): $N = m(g \pm a)$

2.2 Friction & Drag Models

Static: $f_s \le \mu_s N \quad (f_{s,max} = \mu_s N)$

Kinetic: $f_k = \mu_k N$

Rolling: $f_r = \mu_r N$

Angle of Repose: $\tan\theta_{crit} = \mu_s$

Linear Drag: $\mathbf{F}_d = -b\mathbf{v} \Rightarrow v_t = \frac{mg}{b}, \tau = \frac{m}{b}$

Quadratic Drag: $F_d = \frac{1}{2}C_d \rho A v^2 \Rightarrow v_t = \sqrt{\frac{2mg}{C_d \rho A}}$

2.3 Centripetal Dynamics & Banked Roads

$\sum F_c = \frac{m v^2}{r} = m\omega^2 r$

Flat Curve: $v_{max} = \sqrt{\mu_s g r}$

Frictionless Banked Curve: $\tan\theta = \frac{v^2}{gr} \Rightarrow v = \sqrt{gr \tan\theta}$

Banked Curve w/ Friction: $v_{max} = \sqrt{gr\frac{\sin\theta + \mu_s \cos\theta}{\cos\theta - \mu_s \sin\theta}}$

Conical Pendulum: $T\cos\theta = mg, \quad T\sin\theta = \frac{m v^2}{r} \Rightarrow v = \sqrt{gr\tan\theta}, \quad T_{period} = 2\pi\sqrt{\frac{L\cos\theta}{g}}$

Vertical Circle Top: $T_{top} = \frac{mv^2}{r} - mg \quad (v_{min,top} = \sqrt{gr})$

Vertical Circle Bottom: $T_{bot} = \frac{mv^2}{r} + mg$

SECTION 3: WORK, ENERGY, MOMENTUM & COLLISIONS

3.1 Work, Kinetic Energy & Power

$W = \mathbf{F} \cdot \Delta\mathbf{r} = F\,d\cos\theta$

$W = \int_{x_i}^{x_f} F(x)\,dx$

$W = \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{r}$

Spring Work: $W_s = -\frac{1}{2}k(x_f^2 - x_i^2)$

Gravity Work: $W_g = -mg\Delta y$

$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$

Work-KE Theorem: $W_{\text{net}} = \Delta K = K_f - K_i$

$P_{\text{avg}} = \frac{W}{\Delta t}$

$P = \frac{dW}{dt} = \mathbf{F} \cdot \mathbf{v}$

3.2 Potential Energy & Conservation of Mechanical Energy

Conservative Force: $F_x = -\frac{dU}{dx}$

$\mathbf{F} = -\nabla U$

$\Delta U = -\int \mathbf{F} \cdot d\mathbf{r}$

Gravitational PE (local): $U_g = mgy$

Gravitational PE (universal): $U_g = -\frac{G\,M\,m}{r}$

Elastic PE: $U_s = \frac{1}{2}kx^2$

Total Mech Energy: $E_{\text{mech}} = K + U$

Conservation: $W_{\text{nc}} = \Delta E_{\text{mech}} = (K_f + U_f) - (K_i + U_i)$

If $W_{\text{nc}} = 0 \Rightarrow K_i + U_i = K_f + U_f$

3.3 Linear Momentum, Impulse & Center of Mass

$\mathbf{p} = m\mathbf{v}$

$\sum \mathbf{F} = \frac{d\mathbf{p}}{dt}$

Impulse: $\mathbf{J} = \int_{t_i}^{t_f} \mathbf{F}\,dt = \mathbf{F}_{\text{avg}}\,\Delta t = \Delta\mathbf{p} = \mathbf{p}_f - \mathbf{p}_i$

Conservation: $\sum \mathbf{p}_i = \sum \mathbf{p}_f$ (when $\sum \mathbf{F}_{\text{ext}} = \mathbf{0}$)

Center of Mass: $\mathbf{r}_{\text{cm}} = \frac{1}{M}\sum m_i\,\mathbf{r}_i = \frac{1}{M}\int \mathbf{r}\,dm$

Total Momentum: $\mathbf{P} = M\,\mathbf{v}_{\text{cm}}$

$\sum \mathbf{F}_{\text{ext}} = M\,\mathbf{a}_{\text{cm}}$

Rocket Equation: $v_f - v_i = v_{\text{ex}} \ln\frac{M_i}{M_f}$

Rocket Thrust $= v_{\text{ex}}\frac{-dM}{dt}$

3.4 Collisions in One and Two Dimensions

Coefficient of Restitution: $e = \frac{v_{2f} - v_{1f}}{v_{1i} - v_{2i}}$ ($e = 1$: Elastic, $0 < e < 1$: Inelastic, $e = 0$: Completely Inelastic)

Completely Inelastic Final Velocity: $\mathbf{v}_f = \frac{m_1\mathbf{v}_{1i} + m_2\mathbf{v}_{2i}}{m_1 + m_2}$

1D Elastic $v_{1f} = \frac{m_1 - m_2}{m_1 + m_2}v_{1i} + \frac{2m_2}{m_1 + m_2}v_{2i}$

1D Elastic $v_{2f} = \frac{2m_1}{m_1 + m_2}v_{1i} + \frac{m_2 - m_1}{m_1 + m_2}v_{2i}$

Special Case ($m_1 = m_2$): $v_{1f} = v_{2i}$, $v_{2f} = v_{1i}$

SECTION 4: ROTATIONAL DYNAMICS, MOMENT OF INERTIA & ANGULAR MOMENTUM

4.1 Rotational Kinematics & Dynamics

$\omega = \omega_0 + \alpha t$

$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$

$\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t$

$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$

$\omega = \frac{d\theta}{dt}$, $\alpha = \frac{d\omega}{dt} = \omega\frac{d\omega}{d\theta}$

Torque: $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \Rightarrow \tau = r\,F\sin\varphi = F\,d_{\perp}$

$\sum \boldsymbol{\tau}_{\text{ext}} = I\boldsymbol{\alpha} = \frac{d\mathbf{L}}{dt}$

4.2 Moments of Inertia & Structure Theorems

Definition: $I = \sum m_i r_i^2 = \int r^2\,dm = \int \rho\,r^2\,dV$

Parallel-Axis Theorem: $I = I_{\text{cm}} + M\,d^2$

Perpendicular-Axis (Flat Lamina): $I_z = I_x + I_y$

Radius of Gyration: $I = M\,k^2 \Rightarrow k = \sqrt{\frac{I}{M}}$

Hoop / Thin Ring (Axis of symmetry): $I = M\,R^2$

Solid Cylinder / Disk: $I = \frac{1}{2}M\,R^2$

Annular Cylinder (R_1, R_2): $I = \frac{1}{2}M(R_1^2 + R_2^2)$

Solid Sphere (Central axis): $I = \frac{2}{5}M\,R^2$

Thin Spherical Shell: $I = \frac{2}{3}M\,R^2$

Thin Rod (Center): $I = \frac{1}{12}M\,L^2$

Thin Rod (End): $I = \frac{1}{3}M\,L^2$

Rectangular Plate (Sides a, b): $I = \frac{1}{12}M(a^2 + b^2)$

4.3 Rolling Motion, Work-Energy & Angular Momentum

$K_{\text{rot}} = \frac{1}{2}I\omega^2$

$W_{\text{rot}} = \int \tau\,d\theta$

$P_{\text{rot}} = \tau\,\omega$

Rolling w/o Slipping: $v_{\text{cm}} = R\omega$, $a_{\text{cm}} = R\alpha$

Total KE: $K_{\text{tot}} = \frac{1}{2}M\,v_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2 = \frac{1}{2}M\,v_{\text{cm}}^2\left(1 + \frac{I_{\text{cm}}}{M\,R^2}\right)$

Incline Acceleration: $a_{\text{cm}} = \frac{g\sin\theta}{1 + I_{\text{cm}}/(MR^2)}$

Particle: $\mathbf{L} = \mathbf{r} \times \mathbf{p} \Rightarrow L = m\,v\,r\sin\varphi$

Rigid Body: $\mathbf{L} = I\,\boldsymbol{\omega}$

Conservation: $I_i\omega_i = I_f\omega_f$ (when $\sum \tau_{\text{ext}} = 0$)

Gyroscopic Precession: $\Omega_p = \frac{\tau}{L} = \frac{M\,g\,d}{I\,\omega}$

SECTION 5: ELASTICITY, MODULUS OF MATERIALS & STRAIN ENERGY

5.1 Stress, Strain & Moduli Definitions

Tensile/Compressive Stress: $\sigma = \frac{F_\perp}{A}$

Tensile Strain: $\varepsilon = \frac{\Delta L}{L_0}$

Young's Modulus: $Y = \frac{\sigma}{\varepsilon} = \frac{F/A}{\Delta L/L_0} \Rightarrow \Delta L = \frac{F\,L_0}{A\,Y}$

Shear Stress: $\tau = \frac{F_\parallel}{A}$

Shear Strain: $\gamma = \frac{\Delta x}{h} = \tan\theta \approx \theta$

Shear Modulus: $S = G = \frac{\tau}{\gamma} = \frac{F_\parallel/A}{\Delta x/h}$

Bulk Stress: $\Delta P = \frac{F_\perp}{A}$

Bulk Strain: $\frac{\Delta V}{V_0}$

Bulk Modulus: $B = -\frac{\Delta P}{\Delta V/V_0} \Rightarrow \Delta V = -\frac{V_0\,\Delta P}{B}$

Compressibility: $k = \beta = \frac{1}{B} = -\frac{1}{V_0}\frac{\Delta V}{\Delta P}$

Poisson's Ratio: $\nu = -\frac{\varepsilon_{\text{lateral}}}{\varepsilon_{\text{axial}}} = -\frac{\Delta d\,/\,d_0}{\Delta L\,/\,L_0}$

Relations: $Y = 2G(1+\nu) = 3B(1-2\nu)$

5.2 Strain Energy & Spring Analogy

Spring Analogy: $k_{\text{eff}} = \frac{A\,Y}{L_0}$

Elastic Potential Energy: $U = \frac{1}{2}F\,\Delta L = \frac{1}{2}k_{\text{eff}}(\Delta L)^2 = \frac{1}{2}\frac{A\,Y}{L_0}(\Delta L)^2$

Strain Energy Density: $u = \frac{U}{V_0} = \frac{1}{2}\sigma\varepsilon = \frac{\sigma^2}{2Y} = \frac{1}{2}Y\varepsilon^2$

Shear Energy Density: $u_{\text{shear}} = \frac{1}{2}G\gamma^2$

SECTION 6: FLUID MECHANICS: STATICS, BUOYANCY & HYDRODYNAMICS

6.1 Fluid Statics, Pascal's Principle & Manometry

Density: $\rho = \frac{m}{V}$

Specific Gravity: $\text{SG} = \frac{\rho}{\rho_{\text{water}}}$

Pressure: $P = \frac{F}{A}$

Hydrostatic Law: $P = P_0 + \rho g h \Rightarrow \Delta P = \rho g \Delta h$

Pascal's Principle: $\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_2 = F_1\frac{A_2}{A_1}$

U-Tube: $P_1 + \rho_1 g h_1 = P_2 + \rho_2 g h_2$

Barometer: $P_{\text{atm}} = \rho_{\text{Hg}}\,g\,h$

6.2 Archimedes' Principle & Buoyancy

Buoyant Force: $F_B = \rho_{\text{fluid}}\,V_{\text{disp}}\,g$

Floating Body: $\frac{V_{\text{sub}}}{V_{\text{total}}} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$

Apparent Weight: $W_{\text{app}} = W - F_B = (\rho_{\text{obj}} - \rho_{\text{fluid}})V\,g$

Submerged Fraction: %Submerged = SG × 100%

6.3 Fluid Dynamics (Incompressible, Non-Viscous)

Continuity Equation: $Q = A_1 v_1 = A_2 v_2 = \text{const}$

Mass Flow Rate: $\frac{dm}{dt} = \rho\,A\,v = \text{const}$

Bernoulli's Equation: $P_1 + \frac{1}{2}\rho v_1^{\,2} + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^{\,2} + \rho g y_2 = \text{const}$

Torricelli's Law: $v = \sqrt{2gh}$

Venturi Tube: $v_1 = \sqrt{\frac{2\,\Delta P}{\rho[(A_1/A_2)^2 - 1]}}$

Pitot Tube: $v = \sqrt{\frac{2(P_{\text{stag}} - P_{\text{stat}})}{\rho}}$

6.4 Viscosity, Poiseuille Flow & Surface Tension

Viscous Shear: $\frac{F}{A} = \eta\frac{dv}{dy}$

Poiseuille's Law: $Q = \frac{\pi R^4\,\Delta P}{8\,\eta\,L} = \frac{\Delta P}{R_h}$

Hydraulic Resistance: $R_h = \frac{8\,\eta\,L}{\pi R^4}$

Reynolds Number: $\text{Re} = \frac{\rho\,v\,D}{\eta}$ (<2000: Laminar, >4000: Turbulent)

Stokes' Drag (Sphere): $F_d = 6\pi\eta r v$

Stokes' Terminal Velocity: $v_t = \frac{2r^2 g(\rho_{\text{sph}} - \rho_{\text{fl}})}{9\eta}$

Surface Tension: $\gamma = \frac{F}{L}$

Bubble: $\Delta P = \frac{4\gamma}{R}$

Drop: $\Delta P = \frac{2\gamma}{R}$

Capillary Rise: $h = \frac{2\gamma\cos\theta}{\rho\,g\,r}$

SECTION 7: OSCILLATIONS, SIMPLE HARMONIC MOTION & RESONANCE

7.1 SHM Differential Equations & Kinematics

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$x(t) = A\cos(\omega t + \varphi)$$

$$v(t) = -\omega A \sin(\omega t + \varphi)$$

$$a(t) = -\omega^2 A\cos(\omega t + \varphi) = -\omega^2 x(t)$$

$$v_{max} = \omega A$$

$$a_{max} = \omega^2 A$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$v(x) = \pm\omega\sqrt{A^2 - x^2}$$

Total Energy: $$E = \frac{1}{2}kA^2 = \frac{1}{2}m v_{max}^2 = \frac{1}{2}m v^2 + \frac{1}{2}k x^2$$

7.2 Classical Oscillators: Springs & Pendulums

Mass-Spring: $$\omega = \sqrt{\frac{k}{m}} \Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$$

Parallel Springs: $$k_{eq} = \sum k_i$$

Series Springs: $$\frac{1}{k_{eq}} = \sum \frac{1}{k_i}$$

Simple Pendulum: $$\omega = \sqrt{\frac{g}{L}} \Rightarrow T = 2\pi\sqrt{\frac{L}{g}}$$

Physical Pendulum: $$\omega = \sqrt{\frac{mgd}{I}} \Rightarrow T = 2\pi\sqrt{\frac{I}{mgd}}$$

Torsion: $$T = 2\pi\sqrt{\frac{I}{\kappa}}$$

7.3 Superposition, Damped & Driven Oscillations

Superposition (same ω): $$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\delta}$$

$$\tan\alpha = \frac{A_1\sin\varphi_1 + A_2\sin\varphi_2}{A_1\cos\varphi_1 + A_2\cos\varphi_2}$$

Damped Equation: $$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0 \Rightarrow \gamma = \frac{b}{2m}, \quad \omega_0 = \sqrt{\frac{k}{m}}$$

Underdamped: $$x(t) = A_0 e^{-\gamma t}\cos(\omega' t + \varphi), \quad \omega' = \sqrt{\omega_0^2 - \gamma^2} = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

Critical Damping: $$b_{crit} = 2\sqrt{km}$$

Energy Decay: $$E(t) = E_0 e^{-2\gamma t} = E_0 e^{-(b/m)t}$$

Quality Factor: $$Q = \frac{\omega_0}{2\gamma} = \frac{\sqrt{km}}{b}$$

Driven Resonance Amp: $$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (b\omega_d/m)^2}}$$

SECTION 8: MECHANICAL WAVES, SOUND & DOPPLER EFFECT

8.1 Wave Kinematics & Speeds in Media

Wave Equation: $$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

Harmonic: $$y(x,t) = A\sin(kx \mp \omega t + \varphi)$$

$$k = \frac{2\pi}{\lambda}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Wave Speed: $$v = \frac{\omega}{k} = \lambda f = \frac{\lambda}{T}$$

$$v_y = \frac{\partial y}{\partial t} = \mp\omega A\cos(kx \mp \omega t)$$

$$a_y = \frac{\partial^2 y}{\partial t^2} = -\omega^2 y$$

Taut String: $$v = \sqrt{\frac{T}{\mu}} \quad (\mu = \frac{m}{L})$$

Solid Rod: $$v = \sqrt{\frac{Y}{\rho}}$$

Fluid: $$v = \sqrt{\frac{B}{\rho}}$$

Ideal Gas: $$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma R T}{M}}$$

Sound in Air: $$v \approx 331.4 + 0.6\,T_C \text{ m/s}$$

8.2 Power, Intensity & Decibel Level

Average String Power: $$\langle P\rangle = \frac{1}{2}\mu\omega^2 A^2 v$$

Sound Intensity: $$I = \frac{P}{A} = \frac{P}{4\pi r^2}$$

Pressure Intensity: $$I = \frac{\Delta P_{max}^2}{2\rho v} = \frac{1}{2}\rho v\omega^2 s_{max}^2$$

Sound Level (dB): $$\beta = 10\log_{10}\frac{I}{I_0} \quad (I_0 = 10^{-12}\ \text{W/m}^2)$$

Level Difference: $$\Delta\beta = \beta_2 - \beta_1 = 10\log_{10}\frac{I_2}{I_1} = 20\log_{10}\frac{r_1}{r_2}$$

8.3 Standing Waves, Interference, Beats & Doppler Effect

Standing Wave: $$y(x,t) = 2A\sin(kx)\cos(\omega t)$$

Nodes: $$x = \frac{n\lambda}{2}$$, Antinodes: $$x = (2n+1)\frac{\lambda}{4}$$

String & Open Pipe: $$\lambda_n = \frac{2L}{n}, \quad f_n = n\frac{v}{2L} = n f_1 \quad (n = 1, 2, 3, \ldots)$$

Closed Pipe (One End): $$\lambda_n = \frac{4L}{n}, \quad f_n = n\frac{v}{4L} = n f_1 \quad (n = 1, 3, 5, \ldots \text{ odd only})$$

Constructive: $$\Delta r = n\lambda$$, Destructive: $$\Delta r = (n + \frac{1}{2})\lambda$$

Beat Frequency: $$f_{beat} = |f_1 - f_2|$$

Doppler Effect: $$f' = f\left[\frac{v \pm v_O}{v \mp v_S}\right]$$ (Upper signs: Approach • Lower signs: Recede)

Mach Cone: $$\sin\theta = \frac{v}{v_S} = \frac{1}{M}$$

SECTION 9: THERMODYNAMICS, HEAT TRANSFER & ENTROPY

9.1 Temperature, Thermal Expansion & Calorimetry

$T_K = T_C + 273.15$

$T_F = \frac{9}{5} T_C + 32$

Linear: $\Delta L = \alpha L_0 \Delta T$

Area: $\Delta A = 2\alpha A_0 \Delta T$

Volume: $\Delta V = \beta_V V_0 \Delta T \approx 3\alpha V_0 \Delta T$

Thermal Stress: $\frac{F}{A} = Y \alpha \Delta T$

Specific Heat: $Q = m c \Delta T$

Molar Heat: $Q = n C_m \Delta T$

Phase Change: $Q = \pm m L_f$ | $Q = \pm m L_v$

Calorimetry Conservation: $\sum Q_{\text{lost}} + \sum Q_{\text{gained}} = 0$

9.2 Mechanisms of Heat Transfer

Conduction (Fourier): $H = \frac{dQ}{dt} = -k A \frac{dT}{dx} \Rightarrow$ Slab: $H = \frac{k A (T_H - T_C)}{L} = \frac{A(T_H - T_C)}{R_{\text{th}}}$ $(R_{\text{th}} = \frac{L}{k})$

Cylindrical Shell: $H = \frac{2\pi k L (T_1 - T_2)}{\ln(r_2/r_1)}$

Spherical Shell: $H = 4\pi k (T_1 - T_2) \frac{r_1 r_2}{r_2 - r_1}$

Convection (Newton Cooling): $H = h A (T_{\text{surf}} - T_{\text{fluid}})$

Radiation (Stefan-Boltzmann): $H_{\text{emit}} = e \sigma A T^4$

Net Radiation: $H_{\text{net}} = e \sigma A (T^4 - T_{\text{env}}^4)$

9.3 Ideal Gases & Kinetic Theory

Ideal Gas Law: $P V = n R T = N k_B T$ $(R = N_A k_B)$

Kinetic Pressure: $P = \frac{1}{3} \frac{N m}{V} v_{\text{rms}}^2 = \frac{2}{3} \frac{N}{V} \langle K_{\text{trans}} \rangle$

Translational KE / Molecule: $\langle K_{\text{trans}} \rangle = \frac{3}{2} k_B T$

Internal Energy: $U = \frac{d}{2} n R T$ ($d = 3$ Mono, $d = 5$ Dia)

Speeds: $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$, $v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$, $v_{\text{mp}} = \sqrt{\frac{2RT}{M}}$

9.4 First Law & Thermodynamic Processes

First Law: $\Delta U = Q - W$ (W done by system) | $W = \int_{V_i}^{V_f} P \, dV$

Mayer's Relation: $C_p - C_v = R$, $\gamma = \frac{C_p}{C_v}$

Isobaric (P = const): $W = P\Delta V = n R \Delta T$, $Q = n C_p \Delta T$, $\Delta U = n C_v \Delta T$

Isochoric (V = const): $W = 0$, $Q = \Delta U = n C_v \Delta T$

Isothermal (T = const): $\Delta U = 0 \Rightarrow Q = W = n R T \ln \frac{V_f}{V_i} = n R T \ln \frac{P_i}{P_f}$

Adiabatic ($Q = 0$): $\Delta U = -W$, $P V^\gamma = \text{const}$, $T V^{\gamma - 1} = \text{const}$, $W = \frac{P_i V_i - P_f V_f}{\gamma - 1} = \frac{nR(T_i - T_f)}{\gamma - 1}$

Cyclic: $\Delta U_{\text{cycle}} = 0 \Rightarrow Q_{\text{net}} = W_{\text{net}} =$ Area of P-V loop

9.5 Heat Engines, Heat Pumps & Entropy

Thermal Efficiency: $\eta = \frac{W_{\text{net}}}{Q_H} = \frac{Q_H - Q_C}{Q_H} = 1 - \frac{Q_C}{Q_H}$

Carnot Efficiency: $\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$

Refrigerator: $K = \text{COP}_{\text{ref}} = \frac{Q_C}{W} = \frac{Q_C}{Q_H - Q_C} \Rightarrow K_{\text{Carnot}} = \frac{T_C}{T_H - T_C}$

Heat Pump: $\text{COP}_{\text{hp}} = \frac{Q_H}{W} = K + 1 \Rightarrow \text{COP}_{\text{hp,Carnot}} = \frac{T_H}{T_H - T_C}$

Entropy Definition: $\Delta S = \int \frac{dQ_{\text{rev}}}{T}$

Phase Change: $\Delta S = \frac{\pm m L}{T}$

Temperature Change: $\Delta S = m c \ln \frac{T_f}{T_i}$

Ideal Gas General $\Delta S = n C_v \ln \frac{T_f}{T_i} + n R \ln \frac{V_f}{V_i} = n C_p \ln \frac{T_f}{T_i} - n R \ln \frac{P_f}{P_i}$

Free Expansion: $\Delta S_{\text{gas}} = n R \ln \frac{V_f}{V_i} > 0$, $\Delta S_{\text{surr}} = 0$

Second Law: $\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \geq 0$